

Eutectic molecule ligand stabilizes photoactive black phase perovskite

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Formamidinium-rich triiodide perovskites show great promise as light-absorption layers for next photovoltaic technology. However, the intricate process of phase transformation during the crystallization of this perovskite typically results in the presence of undesired hexagonal phases, leading to a degradation in solar cell efficiency and stability. Here we report the use of a hydrogen-bonded eutectic molecule (EM) as a $[\text{PbI}_6]^{4-}$ octahedra ligand to promote the dominant formation of corner-sharing octahedra. Our results demonstrate that the increased ratio of corner-sharing octahedra to face-sharing octahedra can prompt a complete phase transformation and facilitate the formation of pure cubic structure perovskite. Perovskite solar cells based on EM-fabricated films provide a power conversion efficiency of 25.8% (certified 25.7%) and excellent stability with a T_{95} lifetime of ~2,000 h.

Formamidinium (FA)-rich triiodide perovskites have emerged as one of the most promising light-absorber layers in the single-junction and advanced top-cell tandem photovoltaics because of their meticulously engineered electronic bandgap and exceptional photothermal resilience^{1–9}. However, the intricate crystallization of multi-component perovskite prevents the photoactive cubic crystal structure of the α -phase perovskite from residing in an energetically optimized state^{10,11}. Nanoscale phase impurities can easily form in the perovskite because of the incomplete transformation of intermediate phases during the fabrication process or the spontaneous formation of the non-photoactive hexagonal phase¹². These impurities not only act as traps for photoexcited charge carriers but also concurrently serve as the focal point for inducing photochemical

degradation within the absorber layer itself^{13,14}. The phase transition process of the FA-based perovskites contains a series of intermediate phases (that is, face-sharing $[\text{PbI}_6]^{4-}$ octahedra) and generally forms cubic α -phase with corner-sharing $[\text{PbI}_6]^{4-}$ octahedra^{15,16}. Facilitating the formation of pure cubic α -phase and stabilizing the α -phase is of great significance for achieving highly efficient and stable FA-rich triiodide perovskite solar cells (PSCs). Although different additives are developed to passivate defects, suppress ion migration and enhance crystallinity to modulate the growth and phase stability of perovskite films^{4,17–23}, the fundamental strategy for promoting and stabilizing the α -phase remains inconclusive, particularly for FA-rich triple-cation pure iodide perovskites, without compromising the lattice integrity and intrinsic structure.

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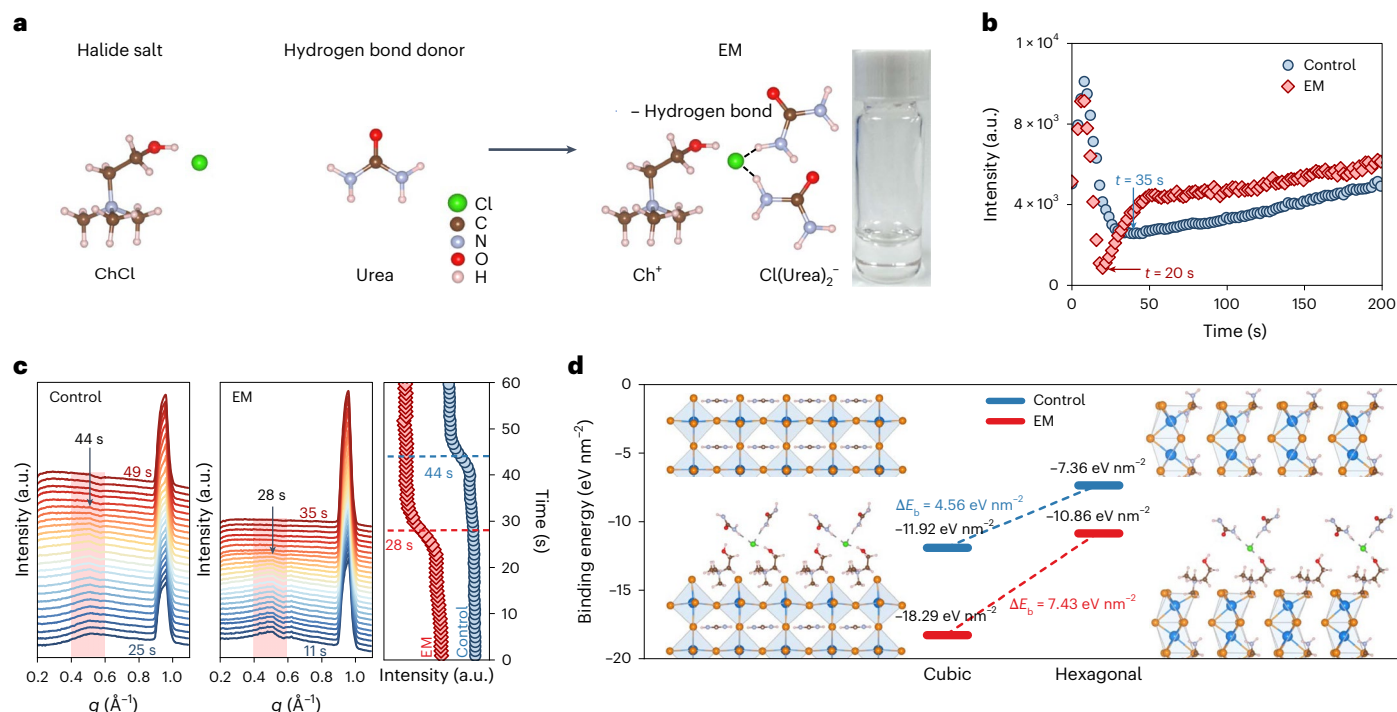


Fig. 1 | EM facilitated phase transition of perovskite. a, Schematic diagram of EM. **b**, In situ PL intensity of the perovskite film during the annealing process. **c**, In situ GIWAXS intensity of the perovskite film during the annealing process. **d**, Binding energy of FA⁺ and EM with the surface of cubic and hexagonal phase perovskite.

Here we have developed a hydrogen-bonded eutectic molecule (EM) with a co-crystal structure composed of choline chloride (ChCl) and carbamide (urea) to modify the phase transition process, eliminate the nanoscale phase impurities and achieve efficient FA-rich triiodide perovskite. This is different from the cations or anions of traditional additives that interact individually, or classic ionic liquids with only binary ion pairing^{24–26}. Our results demonstrate that the hydrogen-bonded EM significantly promotes and stabilizes the α -phase without compromising the intrinsic structure and electrical character. Compared with the face-sharing octahedra of the hexagonal phase, the EM exhibits a stronger interaction with the corner-sharing octahedra of the cubic phase, leading to a decreased formation energy of the cubic α -phase. A complete transformation from the hexagonal to the cubic phase can easily occur during the fabrication of perovskite films, thereby leading to the generation of phase pure perovskite films. Benefiting from the absence of nanoscale phase impurities, the resulting perovskite film exhibits low defect density. Moreover, stable EM can robustly interact with the crystal lattice and further enhance the stability of the resulting perovskite. Finally, the PSC based on EM-fabricated perovskite films exhibits a superior certified power conversion efficiency (PCE) of 25.7% and an excellent ~2,000 h operational stability of T_{95} . Consequently, this work offers a pathway for not only leveraging the richness and adjustability of EM components but also advancing the development of processes to fabricate highly efficient and stable perovskites for various optoelectronic applications.

EM promotes phase pure perovskite formation

Figure 1a illustrates a schematic diagram of the EM synthesis for establishing hydrogen bonding by directly heating two solid reagents of ChCl and urea with a molar ratio of 1:2 at 50 °C for 10 min. The unique EM incorporates both a hydrogen bond donor (urea) and halide salt (ChCl) molecule, which exhibit strong intramolecular hydrogen-bonding interactions between Cl⁻ and urea, forming the Ch⁺–Cl(urea)₂⁻ EM^{24–28}. Fourier-transform infrared spectroscopy and nuclear magnetic resonance results confirm the formation of ChCl–urea EM and hydrogen-bonding interactions, respectively^{25,26} (Supplementary Figs. 1 and 2).

Supplementary Fig. 3 shows that the EM demonstrates exceptional miscibility in dimethylformamide (DMF), which is different from ChCl that is inherently insoluble in DMF. Moreover, the results shown in Supplementary Fig. 4 indicated that Ch⁺ and Cl(urea)₂⁻ of EM remain bound together by ionic bonds rather than dissociate in DMF. The addition of a moderate amount of EM in the precursor has a minimal impact on the average colloidal dimensions, as confirmed by the dynamic light scattering analyses of the perovskite precursor solution (Supplementary Fig. 5). The EM will not cause the precipitation of precursor colloidal and can preserve the uniformity and stability of the precursor colloidal system. It is to be noted that the excessive EM will cause turbidity, indicating the presence of undesired aggregations. X-ray diffraction (XRD) results indicate that the co-precipitated by-products could be attributed to the formation of EMPbI₃, as a result of strong interaction between Ch⁺ of EM and [PbI₆]⁴⁻ octahedra²⁹ (Supplementary Fig. 6). Further experiments with excessive EM addition have demonstrated that EM can spontaneously form low-dimensional perovskite structures at room temperature, in contrast to isolated Ch⁺ (Supplementary Fig. 7).

In situ characterization results indicate that the EM incorporation could modify the phase transition kinetics of FA-rich triiodide perovskites, accelerate the phase transition and eliminate the formation of nanoscale phase impurities. In situ photoluminescence (PL) spectroscopy technique was first used to elucidate the role of EM in fine-tuning phase transition kinetics. Before the anti-solvent, the precursor film began to form in solvent-coordinated complexes and did not exhibit discernible PL behaviour (Supplementary Fig. 8). Upon instant anti-solvent addition, the rapid extraction of the precursor solvent by the anti-solvent expedites the crystallization of perovskite that contains hexagonal intermediate phases with face-sharing [PbI₆]⁴⁻ octahedra, leading to a noticeable onset of PL. During the annealing process, the complete phase transformation of hexagonal to cubic phase with corner-sharing [PbI₆]⁴⁻ octahedra occurs (Supplementary Fig. 9a). The variation in the peak position (Supplementary Fig. 9b) and intensity (Fig. 1b) of the PL indicates that the control films require ~35 s to form the pure cubic phase. With the EM incorporation, the phase transition process is early to ~20 s, indicating that EM significantly

accelerates the formation of the cubic phase. In situ grazing-incidence wide-angle X-ray scattering (GIWAXS) was further used to confirm the phase transition process (Supplementary Fig. 10). During the initial 1 min of the annealing process, the intensity of the intermediate phase in the EM-incorporated film is relatively weak, and its completed disappearance time is shortened by a significant 16 s compared with the control film. These results closely align with the results obtained from in situ PL (Fig. 1c). We also investigate the effect of urea and Ch^+ -only additive (Supplementary Fig. 11), which show negligible and very weak effects on the phase transition, respectively. This indicates that the phase transition kinetics of the EM-fabricated film are unrelated to the urea and are far superior to single Ch^+ . In addition, in situ ultraviolet–visible (UV–vis) absorption spectra during the annealing process also verify the accelerated phase transition process (Supplementary Fig. 12).

Figure 1d indicates that non-volatile EM exhibits strong binding with cubic corner-sharing $[\text{PbI}_6]^{4-}$ octahedra compared with FA^+ , showing a greater difference in binding energy (ΔE_b) with hexagonal face-sharing and cubic corner-sharing $[\text{PbI}_6]^{4-}$ octahedra. The ΔE_b can act as the driving force to maximizing concentrations of corner-sharing $[\text{PbI}_6]^{4-}$ octahedra and further facilitate the formation of stable perovskite cubic phases¹⁵. Moreover, the enhanced hydrogen-bonding interaction between $\text{Cl}(\text{urea})_2^-$ and Ch^+ will make corner-sharing $[\text{PbI}_6]^{4-}$ octahedra more stable (Supplementary Fig. 13). These results further confirm that EM in the precursor solution can significantly facilitate the transition from the hexagonal to the cubic phase. Compared with the single Ch^+ cations, the EM shows a greater ΔE_b (Supplementary Fig. 13), indicating that the EM with hydrogen-bonding structure has more advantages than traditional single ions.

The above comprehensive results confirm the proactive role of EM in the formation of the photoactive cubic phase in perovskite films. Time-of-flight secondary ion mass spectrometry (ToF-SIMS) is further used to investigate the distribution of the components in the EM-incorporated perovskite film. Notably, the chloride (Cl) from methylamine chloride (MACl) will be left behind in the film (Supplementary Fig. 14). Here MACl is not added to the EM-incorporated film. As such, we can confirm that the EM is distributed throughout the perovskite film with all ions having the same distribution (Supplementary Fig. 15). The effect of EM on the performance of the films will be discussed in later part.

High-quality perovskite film enabled by EM

The addition of EM significantly promotes the complete phase transition from the hexagonal to the cubic phase and eliminates the nanoscale phase impurities. The perovskite film fabricated with 5 μl (–1.4 mol%) of EM exhibits remarkable crystalline quality compared with pristine films, showing no discernible impurity hexagonal phases (Fig. 2a,b). According to the results of GIWAXS with different incident angle analyses, the phase distribution throughout the entirety of the EM-incorporated film is remarkably homogeneous. However, hexagonal impurities are observed throughout the entire control film (Supplementary Fig. 16). Similarly, the selected area micro XRD (μXRD) analysis also shows that the control film contains a small amount of hexagonal phase, while the EM-incorporated film shows a pure cubic phase (Supplementary Fig. 17). Scanning electron microscopy analysis reveals that the addition of EM has increased grain size (Supplementary Fig. 18), which is beneficial for reducing grain boundary defects³⁰.

While the EM incorporation improves the crystallinity and phase purity of perovskite, it will not modify optical bandgap and energy level structure. The UV–vis absorption spectra demonstrate that control and EM-fabricated films exhibit nearly identical light absorption and bandgap (Supplementary Fig. 19). The ultraviolet photoelectron spectroscopy (UPS) results indicated that both films exhibit nearly identical work functions and valence band positions, indicating comparable energy level structure (Supplementary Fig. 20). By the outcomes gleaned from X-ray photoelectron spectroscopy (XPS) analyses,

upon meticulous calibration via the carbon spectrum, the resultant spectral profile of lead (Pb) reveals nearly identical binding energies within perovskite films upon the introduction of EM (Supplementary Fig. 21). This demonstrates that neither Ch^+ nor $\text{Cl}(\text{urea})_2^-$ within the EM directly interacts with Pb in the perovskite. Instead, Ch^+ occupies interstice between the corner-sharing $[\text{PbI}_6]^{4-}$ octahedra at the surface as a kind of amine ligand³¹. In all, similar to traditional amine ions, the quaternary alkylammonium Ch^+ from EM can effectively occupy the interstice of $[\text{PbI}_6]^{4-}$ octahedra at the surface of the perovskite without entering into their lattice as a binding ligand³¹. The $\text{Cl}(\text{urea})_2^-$ combined with intramolecular hydrogen bonding enhances the affinity of Ch^+ to the $[\text{PbI}_6]^{4-}$ octahedra and is more energetically beneficial to corner-sharing octahedra. The strong affinity of the hydrogen-bonded EM ligand for the perovskite film is crucial for promoting and stabilizing the cubic phase compared with the single-ion ligand, similar to perovskite nanocrystals^{32,33}.

The EM-fabricated perovskite film also effectively reduces non-radiative recombination within the film. According to PL and PL mapping results, the EM-fabricated film shows a significantly enhanced luminescence intensity and spatial uniformity (Fig. 2c and Supplementary Fig. 22). The time-resolved (TR) PL results show that the PL lifetime (τ) of the EM-incorporated film is much longer (1.47 μs) than that of the control film (0.28 μs) (Fig. 2d and Supplementary Table 1). This notable difference provides evidence for the superior defect passivation effectiveness of EM, significantly attenuating non-radiative recombination within the films. Moreover, transient absorption spectroscopy (TAS) results indicate that the EM-incorporated film exhibits a slower decay in the optical bleaching signal. This is characterized by a faster short-lifetime component and an extended long-lifetime component at 780 nm (Supplementary Fig. 23 and Supplementary Table 2).

EM enhanced efficiency and stability

Benefiting from the advantages of EM, PSCs based on the EM-incorporated film exhibit significantly enhanced performance. Figure 3a compares the current density–voltage (J – V) curves for the small-area PSCs with an area of 0.085 cm^2 . The champion PSC based on the EM-fabricated film achieves a PCE of 25.80% (certified at 25.70% shown in Supplementary Fig. 24), with a short-circuit current density (J_{sc}) of 25.87 mA cm^{-2} , an open-circuit voltage (V_{oc}) of 1.175 V and a fill factor (FF) of 84.88%, which is higher than that of the PSCs based on the control film (23.93%). The external quantum efficiency (EQE)-integrated photocurrent closely aligns with the short-circuit current obtained from the J – V characterization under illumination for the cells (Fig. 3b). The statistical distribution across a sample set of 20 devices also exhibits improved performance and excellent reproducibility (Fig. 3c). Moreover, the PSCs based on the EM-fabricated film also exhibit smaller hysteresis with an average efficiency of 25.26% between forward and reverse scan (Supplementary Fig. 25 and Supplementary Table 3), thereby delivering a stabilized output power of 25.54% (Supplementary Fig. 26). The small hysteresis is likely due to the passivating effect of EM at grain boundaries, effectively mitigating ion migration within the films³⁴. Similarly, as the cell fabrication area scales up to 1 cm^2 , a notable improvement in efficiency is also observed, indicating the excellent uniformity achieved with the EM-incorporated film (Fig. 3d). Furthermore, we verified that EM also enhances the efficiency of PSCs made with FAPbI_3 and $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$ perovskite (Supplementary Fig. 27).

The reduction in non-radiative recombination will notably enhance the V_{oc} of devices. The devices with the EM-incorporated film manifest increased electroluminescence (EL) intensity at equivalent voltages and exhibit higher EQE (EQE_{EL}) values at a current of approximately 25 mA cm^{-2} . This suggests that these devices experience lower V_{oc} losses and yield enhanced performances (Supplementary Fig. 28). Further analysis of the non-radiative recombination contributing

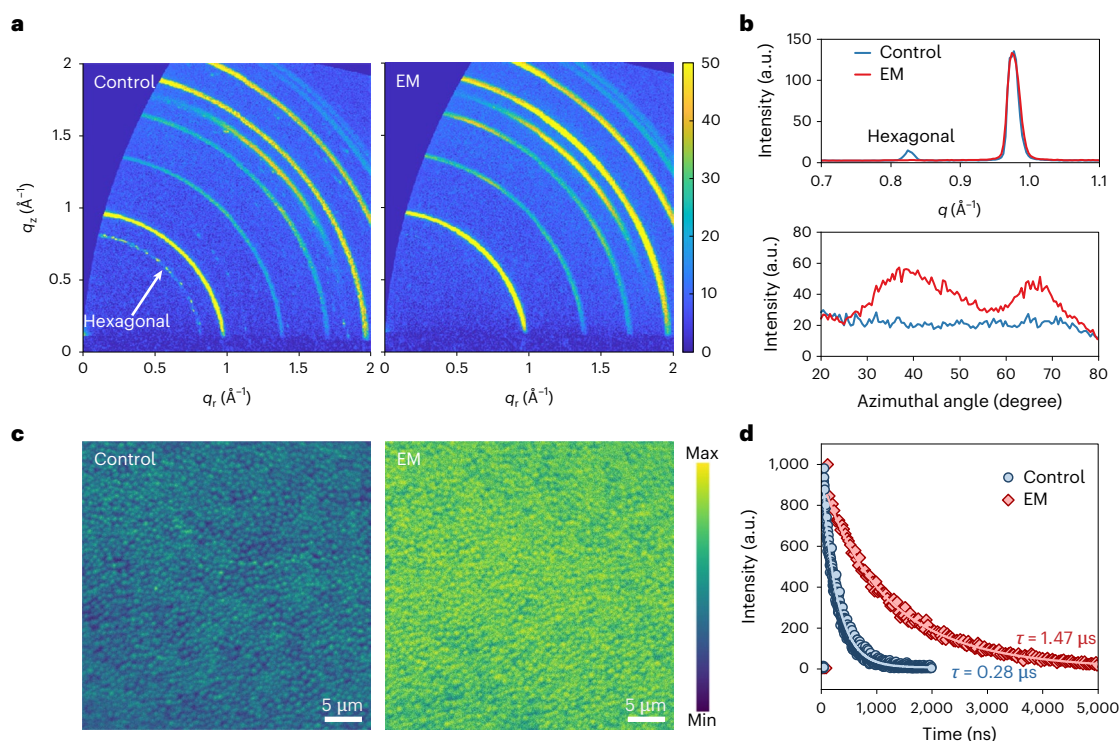


Fig. 2 | Characterization of perovskite film. a, GIWAXS profiles of control and EM-incorporated perovskite film. **b**, The corresponding GIWAXS intensity profiles of control and EM-incorporated perovskite film. **c**, PL mapping image of control and EM-incorporated perovskite film. **d**, TRPL decays of control and EM-incorporated perovskite film.

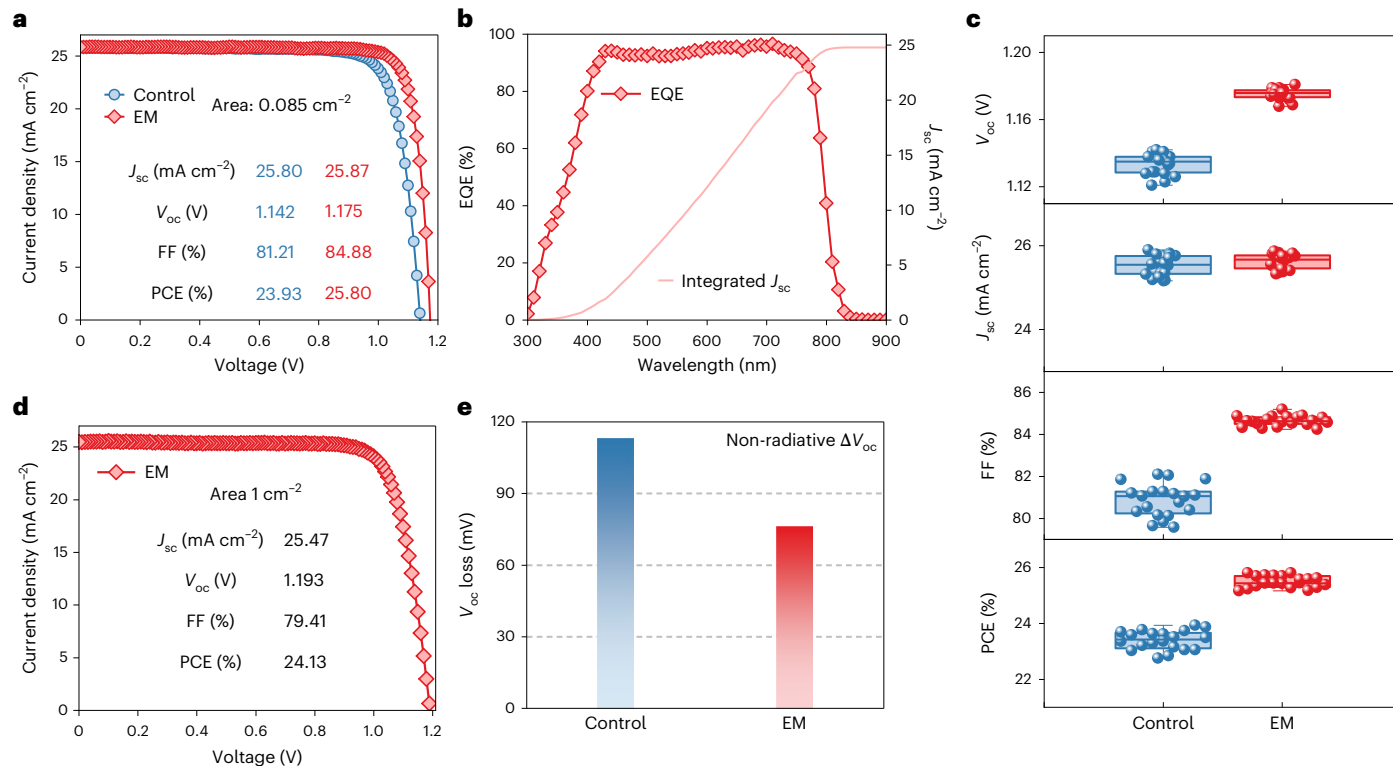


Fig. 3 | Efficiency of PSCs. a, Reverse scan current density–voltage curves of the perovskite photovoltaic devices with a 0.085 cm^2 active area. **b**, EQE spectrum and integrated current of the EM-incorporated PSCs devices. **c**, Photovoltaic parameter distribution of 20 PSCs. The centre line denotes the median value

(50th percentile), while the box contains the 25th to 75th percentiles of dataset. **d**, Reverse scan $J-V$ curve of the EM-incorporated perovskite photovoltaic devices with a 1 cm^2 active area. **e**, Non-radiative V_{oc} loss of photovoltaic devices.

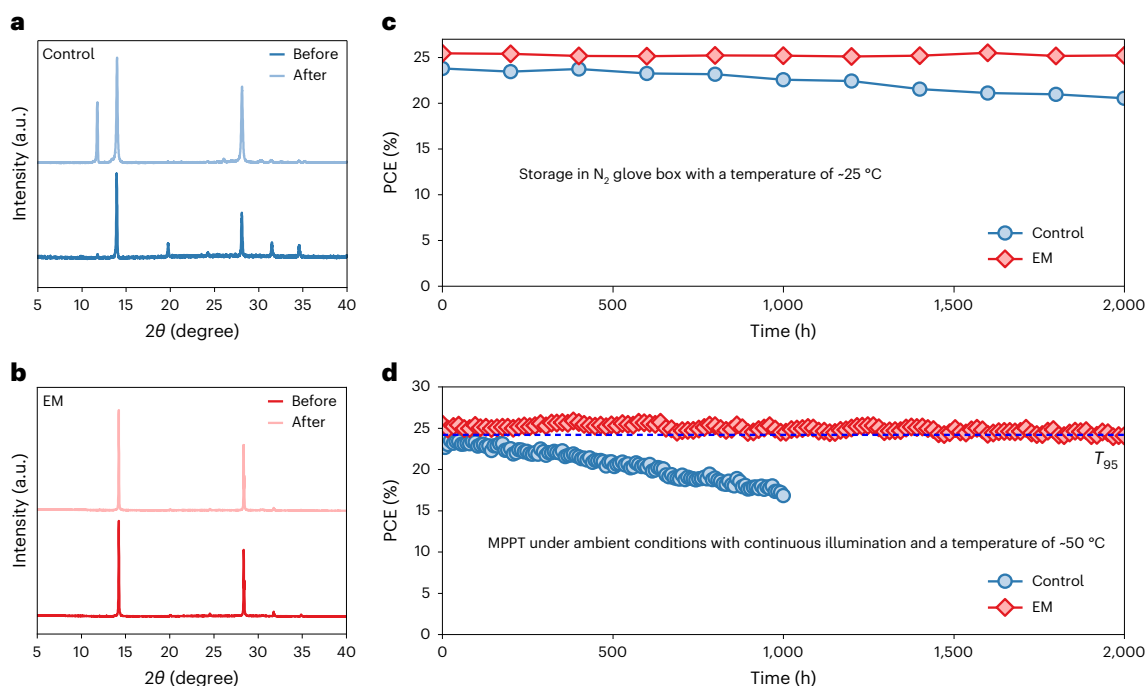


Fig. 4 | Stability of perovskite film and devices. a, XRD patterns of the control perovskite film before and after ageing condition. **b,** XRD patterns of the EM-incorporated perovskite film before and after ageing condition. **c,** PCE during the 2,000 h stability test under storage in glove box condition. The initial efficiencies

are 23.81% for the control device and 25.45% for the EM-incorporated device.

d, PCE during the 2,000 h stability test under maximum power point tracking (MPPT) condition. The initial efficiencies are 23.97% for the control device and 25.48% for the EM-incorporated device.

to V_{oc} loss reveals a significant reduction for the devices with the EM-incorporated film (Fig. 3e).

In addition to improved photovoltaic (PV) performance, the EM-fabricated film exhibits excellent stability. By comparing the XRD results before and after film ageing over 500 h under heating at 85 °C in ambient air (~20% RH), the EM-fabricated film exhibits improved phase stability (Fig. 4a,b). A large number of non-photoactive hexagonal phases are observed in the control film. However, the EM-fabricated film remained pure α -phases. EM effectively suppresses the formation of the hexagonal phase, which is attributed to the EM's adverse effect on the progression of the hexagonal phase. Furthermore, our fabricated devices with the EM incorporation also demonstrated considerably improved storage and operational stability. After 2,000 h of pure storage in the N_2 glove box, the device's stability experiences minimal degradation, while the control sample shows a significant attenuation (Fig. 4c). When tracked at the maximum power point (MPP) with constant illumination at -50 °C, the efficiency of the control device fell by ~30% after 1,000 h, whereas the device fabricated with the EM-incorporated film maintained ~95% of its initial efficiency after 2,000 h (Fig. 4d). Furthermore, the cyclic stability of the EM-fabricated device, simulating a day and night cycle of 12 h of light and 12 h of dark, showed almost no attenuation over 11 days (Supplementary Fig. 29). The increased stability could be attributed to the phase pure $Cs_{0.05}MA_{0.1}FA_{0.85}PbI_3$ perovskite, suppressed phase transition to the hexagonal phase, higher crystallinity and reduced defect density.

Conclusion

In summary, we have demonstrated a novel EM with hydrogen bonding to modify the phase transition process and achieve efficient FA-rich triple-cation perovskite. The perovskite film incorporated with EM shows a facilitating phase transition from hexagonal to cubic phase by enhancing the energy dominance of corner-sharing $[PbI_6]^{4-}$ octahedron formation and eliminating the hexagonal phase impurities. The proposed technology uncovers the microscopic mechanism by

which eutectic additives promote phase transformation and providing an inspiration to explore eutectic system additives with adjustable composition properties.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-024-01596-8>.

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Methods

Materials

All the materials are used as received without further purification. Formamidinium iodide (FAI), methylammonium iodide (MAI) and methylamine hydrochloride (MACl) were bought from Greatcell Solar. Lead iodide (PbI₂) and (2-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-2PACz) were obtained from TCI. Caesium iodide (CsI), C60 and BCP were bought from Xi'an Yuri Solar. Carbamide (urea) and ChCl were bought from Sigma-Aldrich. Acetone, isopropanol (IPA), chlorobenzene (CB), DMF and dimethyl sulfoxide (DMSO) were purchased from J&K.

Solar cell fabrication

ITO glass (15 Ω sq⁻¹; 1.5 × 1.5 cm² for a regular device and 2.5 × 2.5 cm² for a large-area device) was progressively washed by sonication with detergent (Decon 90), deionized water, acetone and IPA for 15 min. The washed ITO glass was dried in a 70 °C oven for over 12 h. Then, the cleaned ITO substrates were subjected to UV ozone for 30 min and transformed into an N₂-filled glove box for film fabrication. Me-2PACz (0.5 mg ml⁻¹ in IPA) was spin-coated onto the cleaned ITO at 3,000 rpm for 30 s, followed by annealing at 100 °C for 10 min. Then, 1.7 M perovskite precursor solutions were constructed by mixing FAI, PbI₂, MAI and CsI in DMF/DMSO mixed solvent (4:1 v/v) with the chemical formula Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃. Then, 10 mol% MACl was added to the precursor solution. For the target perovskite, 5% (v/v) (~1.4 mol%) of the EM based on ChCl and urea was added to the precursor solution, and it was ensured that it was well mixed with the precursor. Then, 50 μl of the prepared precursor solution was spin-coated at 1,000 rpm for 10 s and 5,000 rpm for 30 s onto the Me-2PACz-coated ITO substrate; 180 μl CB as the anti-solvent was dripped on the film at 10 s before the end of the last procedure and then annealed at 100 °C for 30 min. After deposition of the perovskite active layer, 50 μl piperazinium diiodide (0.4 mg ml⁻¹ in IPA) solution was spin-coated onto the film at 5,000 rpm for 30 s and annealed at 100 °C for 5 min. Subsequently, 25 nm C60, 8 nm bathocuproine and 100 nm Cu electrode were sequentially evaporated under a high vacuum (<4 × 10⁻⁶ torr).

PV characterizations

The devices were tested using an AM1.5 G simulator (WAVELABS SINUS-70 AAA solar simulator) in an N₂ glove box. The simulator's power was calibrated to 100 mW cm⁻² by a monocrystalline silicon reference cell associated with a suitable KG1 filter (calibration by Enli Technology, serial number SRC-00406) to keep the spectral mismatch correction at 1.00 ± 0.01. *J*–*V* curves were obtained using a Keithley 2400 source meter by reverse scan (from 1.20 V to –0.1 V) and then forward scan (from –0.10 V to 1.20 V) with a scan rate of 0.1 V s⁻¹, a voltage step of 0.01 V and a delay time of 50 ms. The active area is 0.085 cm² and 1 cm², determined by the metal aperture in front of the device. The EQE measurement was performed by a system combining a xenon lamp monochromatic chopper and a lock-in amplifier with a calibrated silicon photodetector (Hamamatsu mono-Si cell). The steady-state PCE was measured at an MPP and then tested the efficiency. For the stability test, unencapsulated devices were stored in an N₂-filled glove box. *J*–*V* curves and steady-state PCE were regularly tested every 100 h. The long-term operating stability tests were subjected to the protocol of ISOS-L-2/3 (MPP tracking under continuous light and heating) and performed in an N₂-filled chamber under light-emitting diode (LED) illumination. Simultaneous MPP tracking of multiple solar cells was biased at MPP voltages with a custom-built, multi-channel MPP tracking system with high-precision source meters (current resolution of 1 μA; 5 voltage resolution of 1 mV). We calibrated the LED's intensity by adjusting the power when it produced a 1-sun-equivalent current density from the devices. The ambient temperature in the chamber was measured with an infrared thermometer as 40 ± 10 °C during device operation. A silicon photodiode was used to monitor the power change of the lamp, and no apparent changes were observed during MPP tracking.

Other characterizations

Dynamic light scattering was conducted using Zetasizer Nano ZS. The scanning electron microscope images and energy-dispersive X-ray spectroscopy mapping were obtained by Hitachi S-4800 FEG. XRD was taken using a Bruker D2 Phaser with Cu Kα radiation. XPS and UPS were tested on Physical Electronics PHI 5802 with a monochromatic Al Kα X-ray source. ToF-SIMS characterization was performed on ION TOF-SIMS 5. The steady-state PL and time-resolved PL spectra were measured on a PicoQuant FluoTime 300 instrument. A picosecond 375 nm pulse laser (LDH-P-C-75) pulse width of ~40 ps was used to excite the samples. The excitation fluence was ~8 nJ cm⁻². UV–vis absorption spectroscopy measurement was conducted on PerkinElmer LAMBDA 750s. Fourier-transform infrared spectroscopy measurement was conducted on Bruker Vertex 70v. GIWAXS measurements were carried out with a Xeuss 2.0 SAXS/WAXS laboratory beamline using a Cu X-ray source (8.05 keV, 1.54 Å) and a Pilatus3R 300K detector. PL mapping was conducted on a Leica DMI6000b inverted fluorescence motorized phase contrast microscope with a 405 nm laser wavelength. EL was tested on XP-EQE-Adv from Guangzhou Xipu Optics-electrical Technology as LEDs in the dark and under forwarding voltage bias. The femtosecond transient absorption spectra (TAS) were characterized by a set-up based on a regenerative amplified Ti:sapphire laser system from Coherent (400 nm, 35 fs, pump fluence 50 mJ cm⁻²), nonlinear frequency mixing techniques and the Femto-TA100 spectrometer (Time-Tech Spectra).

In situ PL and GIWAXS measurement

For in situ PL measurement, a homebuilt platform consists of a 405 nm laser diode and an Ocean Optics spectrometer (Flame) coupled with optical fibres. The power density of the laser was approximately 100 mW cm⁻² as the intensity reaching the sample could vary by adjusting the set-up. The integration time of each spectrum was 50 ms during the measurement. The synchrotron-based in situ GIWAXS was conducted by the BL02U2 beamline at Shanghai Synchrotron Radiation Facility (SSRF). The corresponding X-ray beam possesses a wavelength of 1.3 Å at a grazing incidence angle of 0.3° and uses an energy of 9.5 keV. The time of each spectrum was 1 s during the measurement. The FIT2D software was used to parse 2D GIWAXS patterns. GIWAXS patterns were presented in scattering vector *q* coordinates based on the equation $q = 4\pi \sin\theta/\lambda$, where θ is half of the diffraction angle and λ is the incident wavelength. In the processing of the GIWAXS data, the vector *q* was calibrated by measuring XRD from a Lanthanum hexaboride reference sample.

DFT calculations

Our calculations were performed using density functional theory (DFT) methods using the Vienna Ab initio Simulation Package (VASP). The electron-ion interactions were described by using the projector augmented wave method. The exchange and correlation energies were determined with the Perdew, Burke and Ernzerhof functional. To accurately describe the dispersion interaction, we use the DFT-D3 method with Becke–Johnson damping for dispersion correction. The geometry optimization was performed until the Hellmann–Feynman force on each atom was smaller than 0.02 eV Å⁻¹. The energy criterion was set to 10⁻⁴ eV in the iterative solution of the Kohn–Sham equation. The Brillouin zone was sampled with 4 × 4 × 1 Monkhorst–Pack *k*-point mesh, and a Gaussian smearing of 0.05 eV was applied to speed up electronic convergence. A vacuum height of 15 Å along the vertical direction was selected to avoid the unwanted interaction between the slab and its period images.

Bi-exponential kinetic fitting model for TRPL and TAS

$$I(t) = A_1 \exp\left(-\frac{t-t_0}{\tau_1}\right) + A_2 \exp\left(-\frac{t-t_0}{\tau_2}\right)$$

where t_0 is the start time of the decay process, τ_1 and τ_2 represent the first- and second-order decay times, and A_1 and A_2 are the weighting coefficients of each decay channel. The average recombination lifetime (τ) is estimated with the A_i and τ_i values from the fitted curve data according to the following equation:

$$\tau = \sum \frac{A_i \tau_i^2}{A_i \tau_i}$$

Non-radiative V_{oc} loss (ΔV_{oc}) calculation

$$\Delta V_{oc} = K_B T / q \ln(EQE_{EL})$$

where K_B is the Boltzmann constant, T is the absolute temperature, q is the elementary charge and EQE_{EL} is the EQE of the EL of the device in the dark.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or the Supplementary Information. Additional information can be obtained from corresponding authors upon reasonable request.

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Author contributions

Z.-W.G., W.C.H.C., B.X. and Y.W. conceived the idea. W.C.H.C. coordinated the work. Y.W., B.X. and W.C.H.C. supported this work. Z.-W.G. fabricated and characterized the devices. J.Z. and J.S. carried out the TAS and PL experiments. G.Z., Y.T. and X.L. contributed to the GIWAXS test and analysis. Z.-W.G., H.S., D.W., J.F. and L.Q. contributed to data analysis. Z.-W.G. wrote the original draft and W.C.H.C., H.S., B.X. and Y.W. reviewed and edited the draft.

Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

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• Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	The tested areas of perovskite solar cells are 0.085cm ² and 1cm ² . (Methods section: PV characterizations) <i>Explain why this information is not reported/not relevant.</i>
Method used to determine the device area	☒ Yes ☐ No	The tested areas of perovskite solar cells are defined by a non-reflective metal mask with aperture areas of 0.085cm ² and 1cm ² . aperture areas of 0.085cm ² and <i>Explain why this information is not reported/not relevant.</i>

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Forward and backward direction J-V scans were performed. (Supplementary Fig. 25)
Voltage scan conditions	☒ Yes ☐ No	All conditions are mentioned in the Methods section. <i>Explain why this information is not reported/not relevant.</i>
Test environment	☒ Yes ☐ No	All perovskite solar cells were tested in N ₂ -filled glove box at room temperature without encapsulated. <i>Explain why this information is not reported/not relevant.</i>
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	<i>Provide a description of the protocol.</i> There is no preconditioning was applied.
Stability of the J-V characteristic	☒ Yes ☐ No	The stabilized power output tracking at the maximum power point was performed. (Supplementary Fig. 26) <i>Explain why this information is not reported/not relevant.</i>

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	There is a very minor hysteresis is observed for target devices herein. <i>Explain why this information is not reported/not relevant.</i>
Related experimental data	☒ Yes ☐ No	J-V plots under forward and reverse scans are shown in Supplementary Fig. 25. <i>Explain why this information is not reported/not relevant.</i>

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	The EQE was performed (Fig. 3b) and experimental details were in Methods section. <i>Explain why this information is not reported/not relevant.</i>
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A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The values of integrated Jsc from EQE and Jsc from the J-V are matched well. <i>Explain why this information is not reported/not relevant.</i>
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	<i>Provide a description of the measurement conditions.</i> There are no tandem solar cells in this work.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	In our lab, the devices were tested using an EnliTech, AAA solar simulator. The light intensity was calibrated using an NREL-calibrated Si solar cell. (Methods section) <i>Explain why this information is not reported/not relevant.</i>
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cell was calibrated and certified from the NREL. (Methods section) <i>Explain why this information is not reported/not relevant.</i>
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	<i>Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements.</i> In our lab, the light spectrum matches well with the reference Si cell and we didn't calculate the spectral mismatch between the reference cell and the tested devices.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	The tested areas of perovskite solar cells are 0.085cm ² and 1cm ² . (Methods section: PV characterizations) <i>Explain why this information is not reported/not relevant.</i>
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	We have provided the certified mask area in Supplementary Fig. 24. <i>Explain why this information is not reported/not relevant.</i>
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	We identify the performance of our cells in Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences (SIMIT). The certified efficiency is 25.7%. <i>Explain why this information is not reported/not relevant.</i>
A copy of any certificate(s)	☒ Yes ☐ No	We have provided the certified efficiency report in Supplementary Fig. 24. <i>Explain why this information is not reported/not relevant.</i>
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	The data from the 20 cells are statistically analyzed and is shown in Fig. 3c. <i>Explain why this information is not reported/not relevant.</i>
Statistical analysis of the device performance	☒ Yes ☐ No	The information is shown in Fig. 3c. <i>Explain why this information is not reported/not relevant.</i>
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	The statement of long-term stability test conditions is in Methods section and the information is shown in Figs. 4c and 4d. <i>Explain why this information is not reported/not relevant.</i>